Turnout Selection and Nonvoter Partisan Preferences

Study brief for ECON 438 Project 1 | US presidential elections 2016 2020 2024

1 Motivation and Research Question

- Election results reveal voters' choices but cannot reveal how abstainers would vote. Turnout selection may therefore distort estimates of the electorate's partisan preferences.
- CES turnout validation allows us to examine this selection and whether the voter–nonvoter gap changed across three presidential elections.

Research question: Among voter-file-matched citizens, how does model-predicted Republican support differ between voters and nonvoters, and how did this gap change from 2016 to 2024?

2 Datasets and Model Variables

Dataset and access	Variables and role
CES Common Content 2016 | 2020 | 2024 About 185,600 respondents before exclusions	Turnout and voter-file match status; reported presidential choice; pre-election preference for validation; age, sex/gender, race/ethnicity, education, income, party identification and ideology; state, year, survey completion and population weights. Use annual vote-validated releases and their guides.
Cost of Voting Index 50 states in each election	FinalCOVI and voting-policy components, merged by state and election year. COVI enters the turnout equation as a proposed exclusion. Check common policy coverage over time and the separate 2020 COVID measure.

Harmonization: Cumulative CES v11 or later provides common coding, but combines unmatched people with no-vote records. Recover match distinctions from annual files; link respondents using year and case ID.

3 Model Setup

Sample and outcomes: Pool matched citizens aged 18+ with complete controls and classified turnout. T = 1 for a validated vote; R = 1 for a reported Republican vote and 0 for another named candidate. Retain nonvoters; audit exclusions of voters with missing choice. Restricting to Republican/Democratic votes is a sensitivity check.

$$T_i = 1[a_i + u_i > 0], \quad a_i = X_i'\alpha + \eta\, COVI_{st} + \mu_s + \tau_t$$
$$R_i = 1[b_i + \varepsilon_i > 0], \quad b_i = X_i'\beta + \delta_s + \kappa_t, \quad Corr(u_i, \varepsilon_i) = \rho$$

X contains pre-election demographics, income, party ID and ideology. Errors are jointly standard normal; state and year effects differ by equation. Estimate weighted full-information selection probit (Stata heckprobit), with appropriate population weights and state-clustered standard errors. Use a robust Wald test of ρ = 0; compare with a voter-only probit.

$$\text{Nonvoter prediction: } \Pr(R_i = 1 \mid T_i = 0, X_i, COVI_{st}) = \Phi_2(b_i, -a_i; -\rho) / \Phi(-a_i)$$

Φ and Φ_2 are the univariate and bivariate normal CDFs. Report weighted voter–nonvoter gaps by year with confidence intervals. Compare with held-out pre-election preferences; add selected demographic × year interactions.

4 Limitations

- Identification depends on COVI affecting turnout but not latent candidate preference. State effects cannot remove changing political confounding; three elections provide limited within-state policy variation.
- Matching errors and the move from Catalist to TargetSmart complicate comparisons. A matched-only baseline excludes many unregistered citizens; compare alternative unmatched classifications.
- Candidate choice is self-reported. Post-wave attrition, candidate nonresponse and third-party exclusions require sensitivity checks; remaining missing choice is assumed ignorable conditional on controls.
- Bivariate normality and stable preferences underpin extrapolation. Estimates describe model-based preferences in the defined sample, not a causal effect of compulsory voting or a definitive full-turnout election result.